Digital Image Processing Project
Spring Term (2026)

Image Processing Project Instructions

Select one of the suggested Projects in the separate List of DIP projects.
You can work in a team of maximum 3 students.

Project will be submitted through two phases as follow:

Phases	**Due Date**	**All Projects**	**Domain specific projects**	**Generic framework Projects**
Phase I	19-Aprial To 23-Aprile	Report with the following sections (Abstract, Problem definition, objectives and Methodology, Results and interpretation) **Running Code of proposed methodology**	known as pre-processing phase: **Depending on the project you choose** Use any image processing enhancing techniques (e.g. histogram enhancement, contrast stretching, noise removal, filtration, …etc). Also, should apply Frequency domain with it's filter to finalize pre-processing phase.	**Depending on the project you choose** Implement one enhancing algorithm (not allowed to use built in function) (e.g. you cannot use histeq function to implement histogram equalization)
Phase II	12- May To 16- May	**Running Code of proposed methodology** **Power point presentation (10 Minutes)**	Use any image processing segmentation/feature extraction method. Use any image processing classification/clustering method.	**Depending on the project you choose** Implement segmentation or feature extraction algorithm (Not allowed to use built in function. Implement classification or clustering algorithm. (Not allowed to use built in function).

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Phase I (15 Marks)

Check the above requirements.

Phase II (30 marks)

Every group should prepare documentation as research paper with good dataset.

All phases of machine learning should represent

- Good Title
- Your name
- Abstract (Max 250 -300 word)
- Introduction: write good introduction with the problem and the main contribution in this work
- Related work (at least 15 studies).
- Methodology and dataset description (write in this section the main methodology that bring good accuracy with you in your dataset).
    - Data set description, collection, and capturing
    - Preprocessing steps
    - Feature extraction using (learning features from CNN models, and feature engineering features (LBP, GLCM, SURF, SIFT,…..etc.))
    - Feature fusion between these all features.
    - Feature reduction and selection.
- Proposed Model: You should draw good and attaractive model with all phases you applied. Also discuss each phase in 2 or 3 statements.
- Results and discussion: In this section should interpret well the results you have with figures and images from the results for all steps (preprocessing- feature extraction, feature reduction and selection).

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- Do comparisons for different feature extraction using different architecures from CNN method (such as: VGG16, VGG19, EfficientNet,Resnet,…..etc.).
- Merge the best architecture features with the feature engineering features.
- Apply and compare different machine earning methods using (accuracy, sensitivity, specifity, precision, recall, F-measure, AUC-ROC, kappa coefficient). Good interpretation and make **tables and figures** for the results.
- Conclusion and future work
- Refrences